

Business Requirement Document (BRD) UPI Analytics Platform using Databricks & Power BI

1. Project Overview

The objective of this project is to build a scalable and real-time UPI Analytics Platform using Medallion Architecture on Databricks and interactive reporting using Power BI. The platform will ingest data from multiple sources including UPI transaction gateways, bank settlement systems, merchant systems, customer master data, fraud detection systems, and device logs.

2. Business Objectives

- Monitor real-time UPI transactions and settlement activities.
 - Generate executive dashboards and operational reports.
 - Improve fraud detection and risk monitoring capabilities.
 - Provide merchant and customer analytics.
 - Enable faster business decision-making using centralized analytics.
 - Build scalable and governed data architecture using Delta Lake.
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3. Scope of the Project

In Scope:

- Data ingestion from multiple structured and semi-structured sources.
- Bronze, Silver, and Gold layer implementation in Databricks.
- KPI aggregation and reporting datasets.
- Power BI dashboard development.
- Real-time and batch data processing.

Out of Scope:

- Core banking transaction processing.
 - Payment gateway development.
 - Third-party settlement system modifications.
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4. Source Systems

- UPI Transaction Gateway
 - Bank Settlement Systems
 - Merchant Systems
 - Customer Master Data
 - Fraud Detection Logs
 - Device and Application Logs
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5. Proposed Architecture

The solution will use Medallion Architecture in Databricks consisting of Bronze, Silver, and Gold layers.

Bronze Layer: Raw ingestion of source data.

Silver Layer: Cleansed, transformed, and enriched datasets.

Gold Layer: Aggregated business KPIs optimized for reporting and analytics.

6. Functional Requirements

- Ingest CSV and JSON files into Delta Lake.
 - Support schema evolution and data validation.
 - Perform data cleansing and deduplication.
 - Generate KPI aggregation tables.
 - Build Power BI dashboards for multiple business teams.
 - Support real-time streaming analytics.
 - Implement fraud detection indicators and alerts.
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7. Non-Functional Requirements

- Scalability to handle millions of UPI transactions daily.
 - High availability and fault tolerance.
 - Secure data access using RBAC and ACLs.
 - Data lineage and governance support.
 - Low latency dashboard refresh capability.
 - Audit and compliance readiness.
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8. Data Layers and Tables

Bronze Tables:

- bronze_upi_transactions
- bronze_bank_settlement
- bronze_fraud_logs

Silver Tables:

- silver_upi_transactions
- silver_fraud_analysis

Gold Tables:

- gold_bank_kpi
 - gold_merchant_kpi
 - gold_fraud_monitoring
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9. Power BI Reports and Dashboards

- Executive Summary Dashboard
- Transaction Analysis Report
- Bank Performance Dashboard
- Merchant Analytics Dashboard
- Customer Behavior Report
- Fraud Detection Dashboard

- Settlement Reconciliation Dashboard
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10. Key KPIs

- Total Transactions
 - Total Transaction Amount
 - Success Rate
 - Failed Transactions
 - Average Transaction Amount
 - Active Customers
 - Active Merchants
 - Fraud Count
 - Settlement Success Rate
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11. Benefits of the Solution

- Centralized enterprise analytics platform.
 - Faster reporting and decision-making.
 - Improved fraud detection capabilities.
 - Real-time operational monitoring.
 - Better customer and merchant insights.
 - Optimized data processing using Delta Lake.
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12. Technology Stack

- Databricks
 - Delta Lake
 - PySpark
 - Azure Data Factory / Auto Loader
 - Azure Event Hub / Kafka
 - Power BI
 - SQL
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13. Security and Governance

- Role-based access control (RBAC).
 - Unity Catalog integration.
 - Data masking for sensitive fields.
 - Audit logging and lineage tracking.
 - Secure API and storage access.
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14. Expected Business Outcomes

- Improved UPI operational visibility.
- Enhanced fraud monitoring and prevention.

- Better merchant and customer engagement.
 - Faster executive reporting.
 - Scalable analytics platform for future growth.
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